

Alberto Lillo

Robotics & AI Engineer
PhD in Human-Robot Interaction

Contact

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About Me

PhD-trained roboticist and AI engineer building embodied cognitive architectures for social and humanoid robots: multimodal perception, LLM/VLM-based reasoning, task planning, memory, user modelling, dialogue and closed-loop robot action through ROS/ROS2. Experienced in deploying and evaluating adaptive behaviours on physical robots, in the lab and at public events.

Robotic Platforms

Booster K1 (humanoid) · iCub (humanoid) · NAO · Pepper · Reachy Mini · Sanbot Elf · Temi

Publications

15 publications, incl. IEEE Transactions on Affective Computing, ACM/IEEE HRI, IEEE RO-MAN, ACM ICMI, ACM UMAP.
Full list on Google Scholar

Live Demos & Outreach

A&T – Automation & Testing, Turin · 2 editions
Maker Faire Rome · 2025
European Researchers' Night, Turin · 4 editions
Live robot demos for the department's third mission.

Languages

Italian · English

Experience

Research Fellow – Feel Even Better project Jun 2026 – present
Dept. of Computer Science, University of Turin · funded by Intesa Sanpaolo Innovation Center
- Social and assistive robotics for people with autism: adaptive, personalised robot behaviours.

Research Fellow – Robots for accessible museums Nov 2025 – Apr 2026
Dept. of Computer Science, University of Turin · funded by University of Turin
- Social and telepresence robots for accessibility and inclusion in small museums.

Visiting Researcher – Cognitive Robotics Lab 2024 – 2025
University of Manchester, UK · Prof. Angelo Cangelosi · two stays: Jan 2024 – Jun 2025, Oct – Nov 2025
- Cognitive architectures for HRI: perception → understanding → appraisal → memory → decision → action.

Visiting Researcher – CONTACT Unit Oct – Dec 2023
Istituto Italiano di Tecnologia (IIT), Genoa · Dr. Alessandra Sciutti
- Cognitive architectures for human-robot interaction.

Research Fellow – Wolly educational robot 2021 – 2022
Dept. of Computer Science, University of Turin · Google Educator Grant
- Affective, adaptive robot companion for children with disabilities in educational settings.

Education

PhD in Computer Science – Social and Assistive Robotics 2022 – 2026
University of Turin · Supervisors: Prof. Cristina Gena, Dr. Alessandra Sciutti (IIT)
- Empathic, adaptive cognitive architectures for social robots (PNRR doctoral scholarship).
- **Don't cHRI**: multimodal emotion recognition and emotional alignment of robot behaviour, published in IEEE Trans. Affective Computing and RO-MAN 2025.

MSc in Artificial Intelligence and Information Systems 2019 – 2022
University of Turin · Thesis: Emotion and Face Recognition in Robot Interaction

BSc in Computer Science 2016 – 2019
University of Turin · Thesis: Emotional AI and Social Robots: the Sanbot case

Teaching & Supervision

- Teaching assistant, MSc courses in Human-Computer Interaction and Web Technologies (40 h each), University of Turin, 2024 – 2026.
- Co-supervisor of 8 BSc and MSc theses on social robotics, HRI and deep learning for robot perception.

Skills

- **Robotics:** ROS/ROS2, robot middleware, SLAM, perception pipelines, sensor integration and fusion
- **AI:** LLM/VLM integration, agentic pipelines, computer vision (YOLO), speech (Whisper, STT/TTS), multimodal perception, affective computing, user modelling, dialogue systems
- **Programming:** Python, C++, C, Java, JavaScript · PyTorch, TensorFlow
- **Systems:** Docker, CI/CD, HPC pipelines, module integration · experimental design and human-subject evaluation